

Inspironics Onboarding

From Factory Floor to Cloud Intelligence



D e s i g n

B u i l d

S e c u r e C o m m i s s i o n O p e r a t e

DESIGN

BUILD

SECURE

COMMISSION

OPERATE

OPTIMIZE

SCALE

Executive Autonomous

C X

C K

C D

C E

Four Architecture

CX (Hospitality) · CK (IoT PaaS)
· CD (Guest Engagement) · CE
(ESG & Cognitive Enterprise)

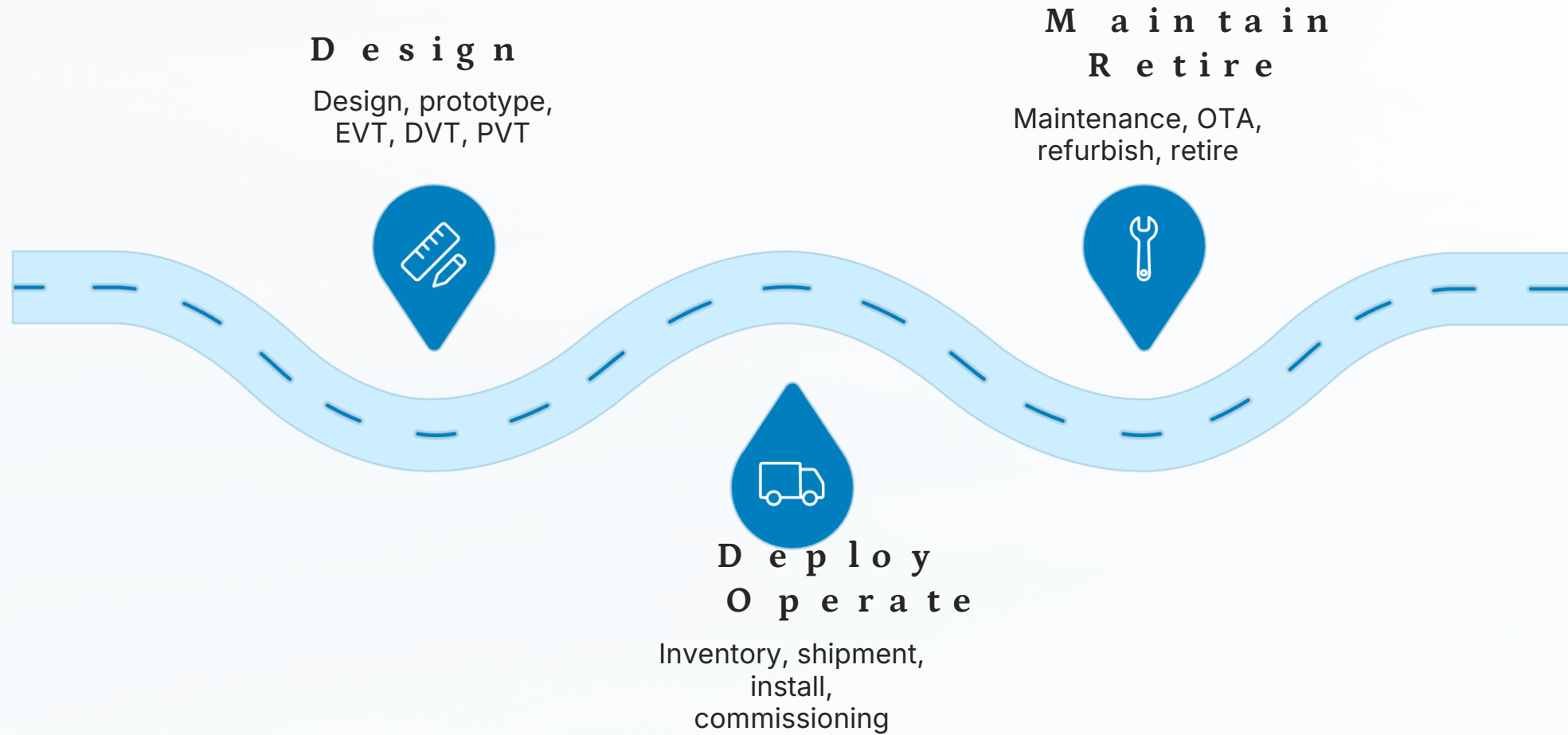
Core

Secure-by-design · Zero-touch
provisioning · Multi-cloud
readiness · Digital Twin
enablement · Agentic AI
operations

End-to-End

Every device — from factory floor to retirement — is traceable, auditable, and AI-ready across its full lifecycle

E n-tōE n d



Every state change becomes an **auditable, immutable lifecycle event** — ensuring full traceability from silicon to retirement.

P r o d u c t

P r o d u c t

F i r m w a r e

D e v i c e

A s s e t

C u s t o m e r

O p e r a t i o n s

U p s t r e a m

- Product Definition & BOM Management
- Firmware Management & versioning
- Manufacturing Records & traceability

D o w n s t r e a m

- QA Validation & Security Certification
- Deployment Governance
- End-of-Life Management

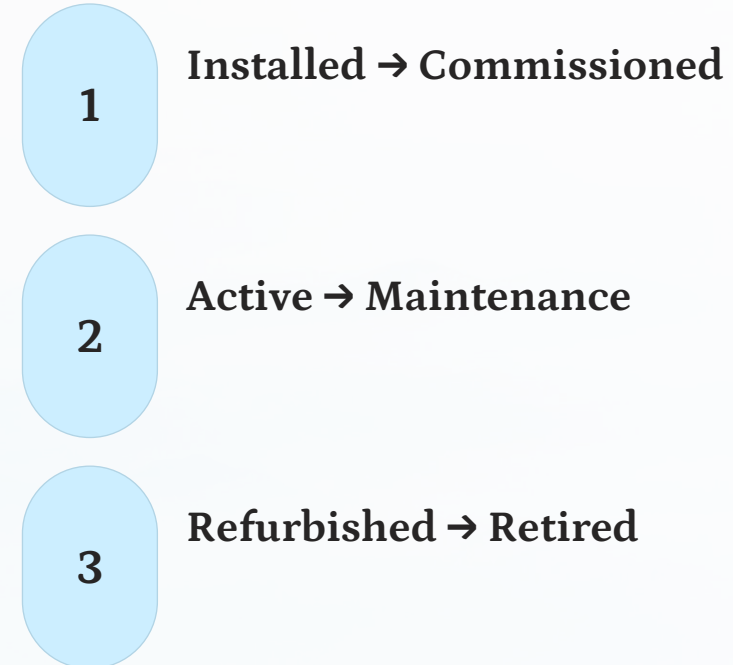
 Continuous traceability: **Product** → **Firmware** → **Device** → **Asset** → **Customer** → **Operations**

Device

Pre-Deployment



Operational



Color-coded state transitions enforce governance and prevent unauthorized state jumps

D e v i c e

H a r d w a r e



U U I D ,
M A C



H a r d w a r e
M a n u f a c t u r i n



F a c t o r y

S e c u r i t y



D e v i c e
P u b l i c



S e c u r e



W a r r a n t y



Every physical device is bound 1:1 to a Digital Twin — no device exists without a cryptographic identity

F a c t o r y

A s s e m b l y

C a l i b r a t i o n

T e s t i n g

S e c u r i t y

T e s t i n g

Smart factory production lines
generate **immutable manufacturing
records** at every stage

F u n c t i o n a l

S e c u r i t y

E n v i r o n m e n t

P e r f o r m a n c e

C o n n e c t i v i t y

S e c u r e



T P M
E l e m e n t



S e c u r e



S i g n e d



O T A

P K I



F a c t o r y

Certificate issuance at production



D e v i c e

Cryptographic binding at hardware level



M u t u a l

Verified handshake before any connection

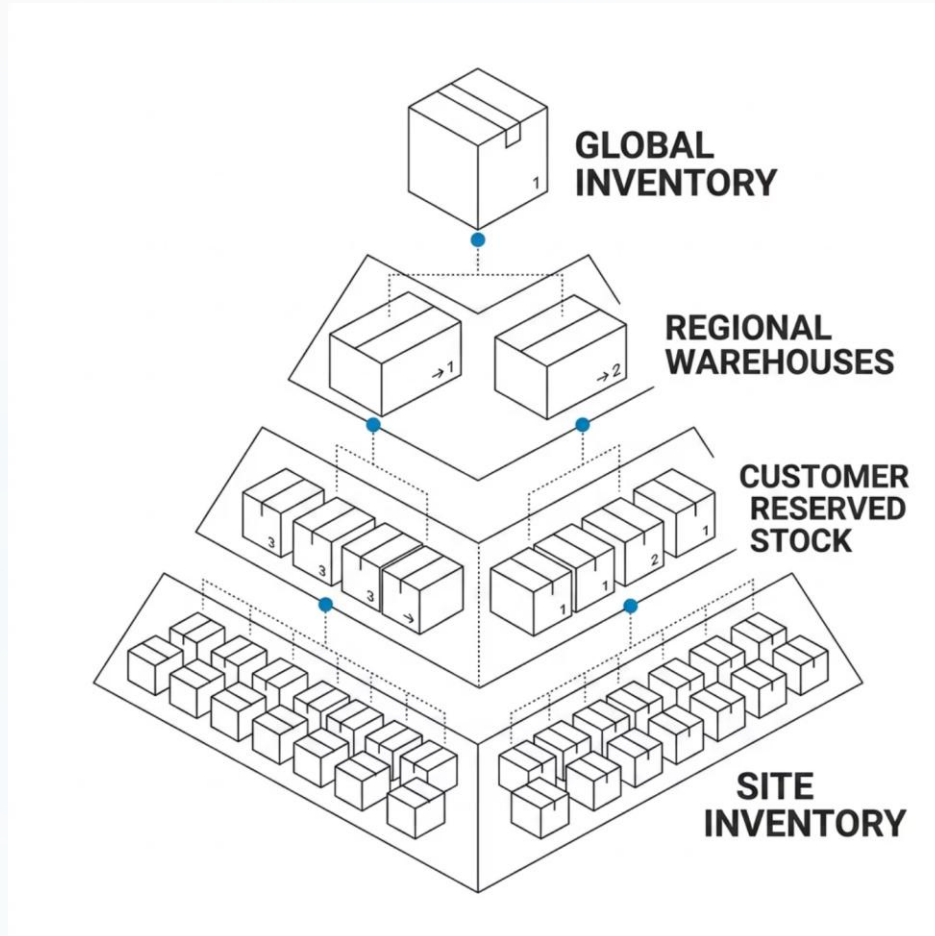


C l o u d

Secure enrollment into platform registry

I n v e n t o r y

W a r e h o u s e



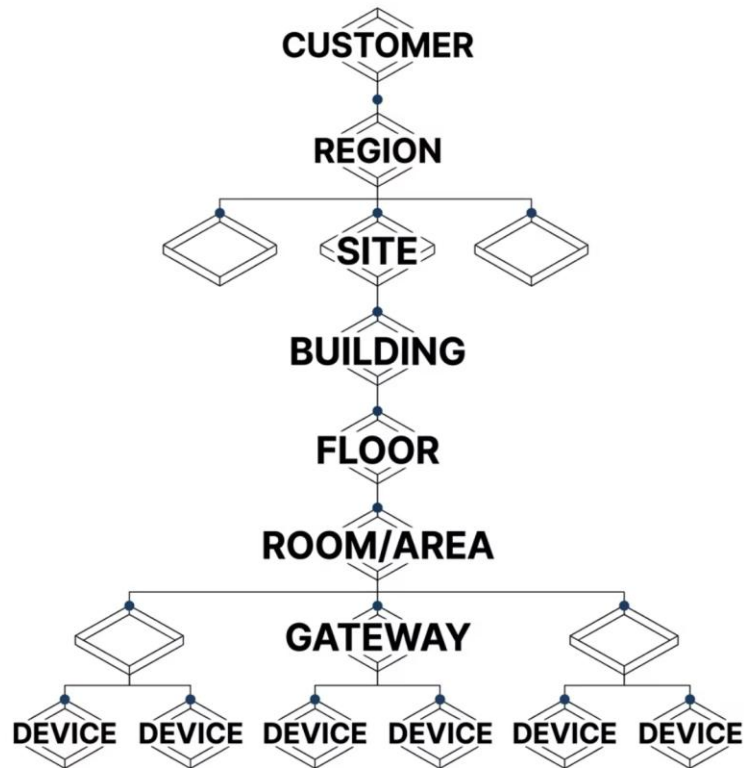
T r a c k e d

- Devices & Gateways
- Spare Parts & Accessories
- Warranty records
- Shipment manifests



Digital supply chain dashboard provides **real-time visibility** across all inventory tiers — no blind spots, no surprises

C u s t o m e r



P l a t f o r m s p e c i f i c

Hierarchical model enables granular access control, billing, and reporting at every level

C X-H o s p i t a l i t y

Hotel → Floor → Room

C E-I n d u s t r i a l

Plant → Line → Asset

C K-I o T

Campus → Zone → Sensor Cluster

I n s t a l l a t i o n

01

A r r i v e

02

S c a n

03

G P S

04

N e t w o r k

05

S i g n a l

06

S i t e

07

C o m m i s s i o n i n g



D e v i c e

QR Code Scan, GPS Capture,
Photo Documentation —
instant device identification in
the field



N e t w o r k

Device Discovery, Signal
Quality Assessment, and
connectivity verification before
proceeding



S i t e

Site Mapping, Commissioning
Checklist — tablet and phone
workflows eliminate all paper-
based records

C o m m i s s i o n i n g

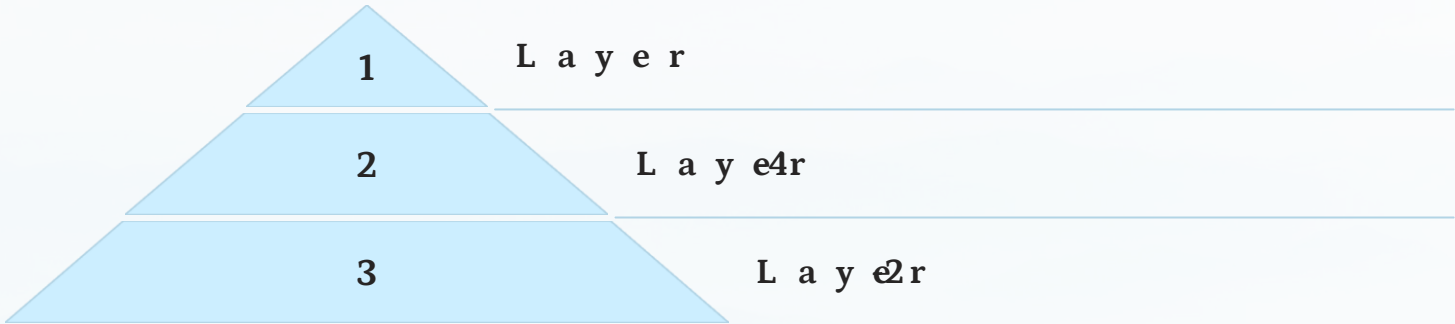


I n s t a l l a t i o n N e t w o r k C e r t i f i c a t e C l o u d S e n s o r
V a l i d a t i o n R e g i s t r a t i o n V a l i d a t i o n

✓ C o n n e c t i v i t y	✓ T e l e m e t r y	✓ A l a r m s
✓ C o m m a n d s	✓ R e p o r t s	

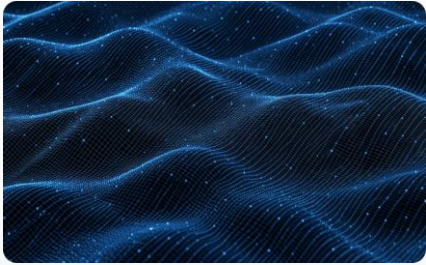
No device goes live without completing **every** validation gate

E n t e r p r i s e



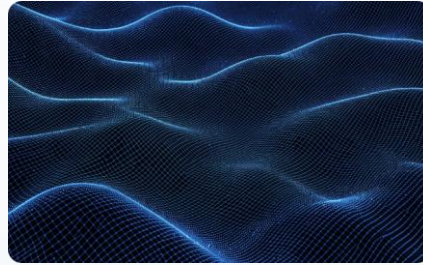
1	L a y e r Exclusive OT WiFi with VLAN Segmentation — IoT traffic fully isolated from corporate IT
2	L a y e r Gateway Network with MQTT Transport — lightweight, reliable pub/sub messaging at scale
3	L a y e r Multi-cloud ingestion, stream processing, analytics, and Digital Twin synchronization

N e t w o r k



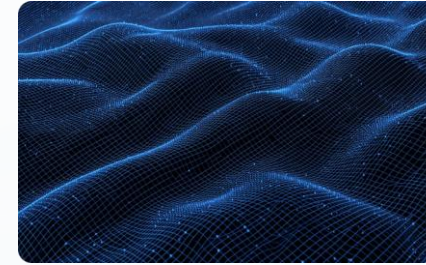
S t a r

Hospitality & Commercial Buildings — simple, low-cost, easy to support and troubleshoot



H u b - S p o k e

Enterprise Multi-Site — centralized governance, unified policy control across locations



M e s h

Agriculture, Utilities, Smart Cities — self-healing, extended coverage, resilient by design

E d g e

G a t e w a y



P r o t o c o l



L o c a l



L o c a l



M Q T T



O T A



D i g i t a l

S u p p o r t e d

B A C n e t

M o d b u s

O P-~~U~~ A

M Q T T

R E S T

- Gateways act as the **intelligent edge** — processing locally, syncing globally:
Sensor → Gateway → Enterprise

Secure

Encryption

T L S

M Q T T S

H T T P S

M u t u a l

O A u t h 2

J W T

Defense-in-Depth

1 Device

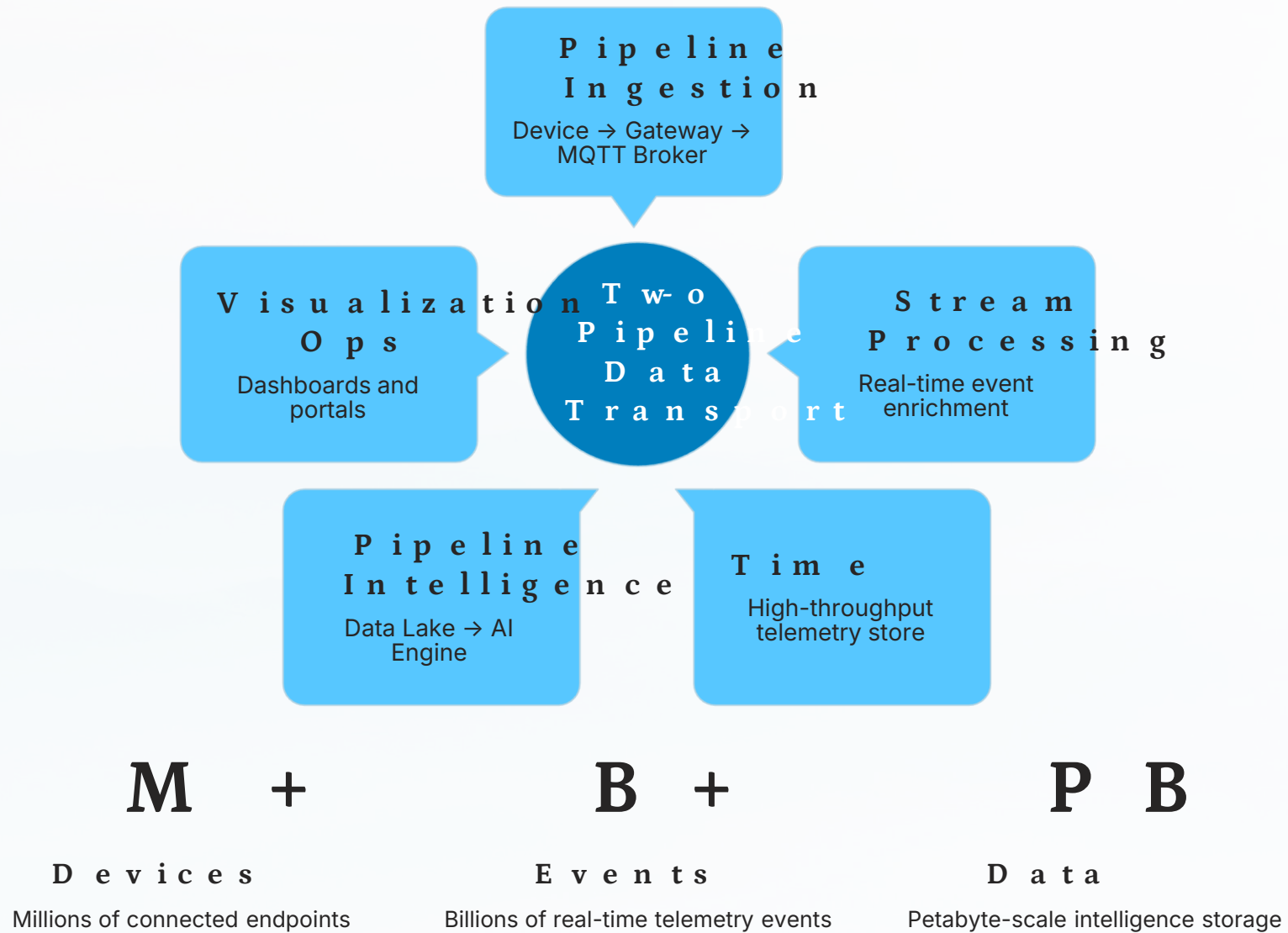
2 Network

3 Application

4 Cloud

✓ Every communication channel is encrypted, authenticated, and auditable end-to-end

D a t a



M u-Cloud



S u p p o r t e d

- AWS IoT Core — device registry, rules engine, event routing
- Azure IoT Hub — digital twin, stream analytics, RBAC
- Google Cloud IoT Ecosystem — pub/sub, BigQuery, AI/ML pipelines

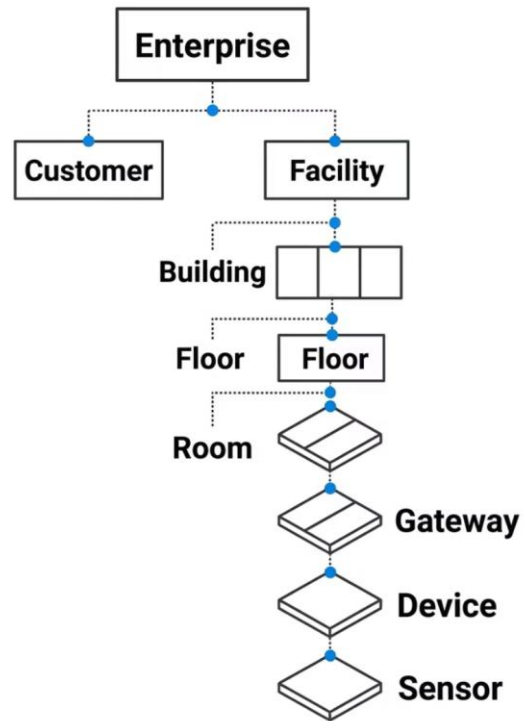
C o r e

- Device Registry & Digital Twin
- Rules Engine & Event Processing
- Analytics & AI Orchestration

- ☐ Cloud-agnostic design prevents vendor lock-in and enables hybrid deployment strategies

D i g i t a l

T w i n



W h a t

- Real-time state & live telemetry
- Full historical data & event log
- Predictive analytics & anomaly forecasts

B u s i n e s s

P r e d i c t i v e

Prevent failures before they occur

E n e r g y

AI-driven consumption reduction

A u t o n o m o u s

Self-managing facility intelligence

M o b i l e



I n s t a l l e r

Device onboarding, QR scan



T e c h n i c i a n

Diagnostics, commissioning



F a c i l i t y

Monitoring, work orders,
predictive maintenance



O p e r a t i o n s

Portfolio analytics, AI
recommendations



E x e c u t i v e

ESG reporting, C-suite dashboards

O n b o a r d i n g

M o n i t o r i n g

D i a g n o s t i c s

W o r k

P r e d i c t i v e

E S G

Single mobile platform serves every stakeholder — from field technician to C-suite

Device



Provisioning
Configuration



OTA
Updates



Certificate
Rotation



Health
Monitoring



Intelligent
Alerting



Anomaly
Detection



Lifecycle

Lifecycle

- Provisioning & Configuration management
- OTA Firmware Updates at scale
- Certificate Rotation & renewal
- Health Monitoring & diagnostics

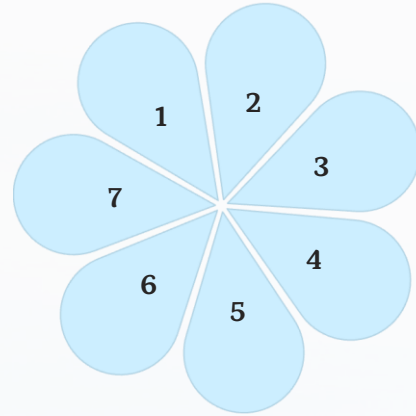
Operational

- Intelligent Alerting & escalation
- Anomaly Detection via AI
- Lifecycle State Management



Single Pane of Glass — unified visibility and control across every device, site, and cloud

A I



- 1 A I**
Central intelligence orchestrating autonomous operations
- 2 D e v i c e**
Edge-level anomaly detection and local decisions
- 3 G a t e w a y**
Protocol handling and device coordination
- 4 F a c i l i t y**
Facility control and automation optimization
- 5 E n e r g y**
HVAC and energy system optimization
- 6 M a i n t e n a n c e**
Predictive maintenance and root cause analysis
- 7 E S G**
Sustainability compliance reporting



D e v i c e

Edge-level intelligence for local anomaly detection, protocol handling, and autonomous decision-making at the hardware layer



F a c i l i t y

Continuous optimization of HVAC, lighting, and energy systems — autonomous actions without human intervention



M a i n t e n a n c e

Predictive maintenance scheduling, root cause analysis, and sustainability compliance reporting — fully automated



Agents operate continuously — **detecting, deciding, and acting** without human intervention

P r o d u c t

C ~~X~~ H o s p i t a l i t y

- Room automation & control
- Guest experience personalization
- Energy management & reporting

C ~~D~~ G u e s t

- Personalization at scale
- Occupancy intelligence
- Service orchestration

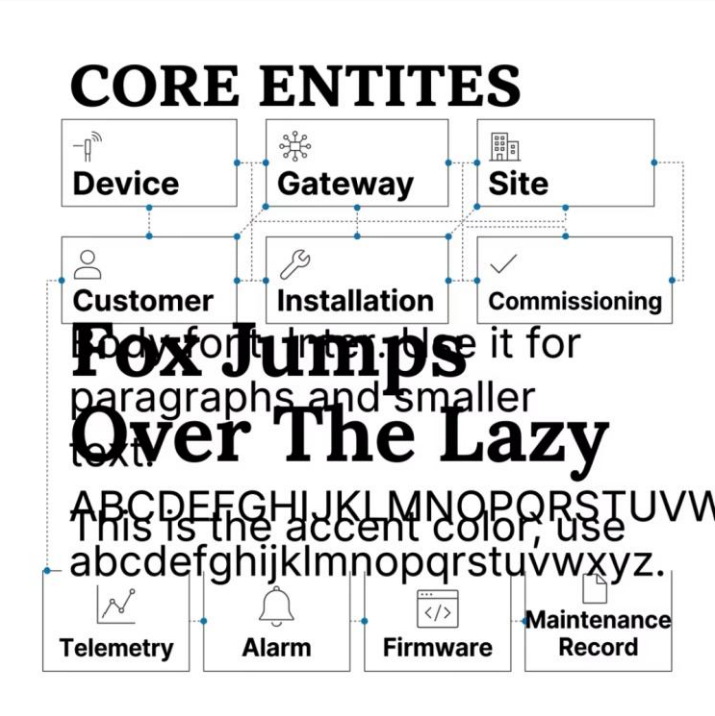
C ~~K~~ I o T

- Device management backbone
- Connectivity fabric
- Data ingestion & routing

C ~~E~~ C o g n i t i v e

- ESG sustainability reporting
- AI-driven facility intelligence
- Enterprise-wide orchestration

E n g i n e e r i n g



C o r e O p e r a t i o n a l

ER model enforces **referential integrity** across the full device lifecycle — factory record to retirement log

D e v i c e

G a t e w a y

S i t e

C u s t o m e r

I n s t a l l a t i o n

C o m m i s s i o n i n g

T e l e m e t r y

A l a r m

F i r m w a r e

M a i n t e n a n c e

D e v S e c O p s



G i t H u b J e n k i n s S e c u r i t y C o n t a i n e r D e p l o y m e n t
B u i l d

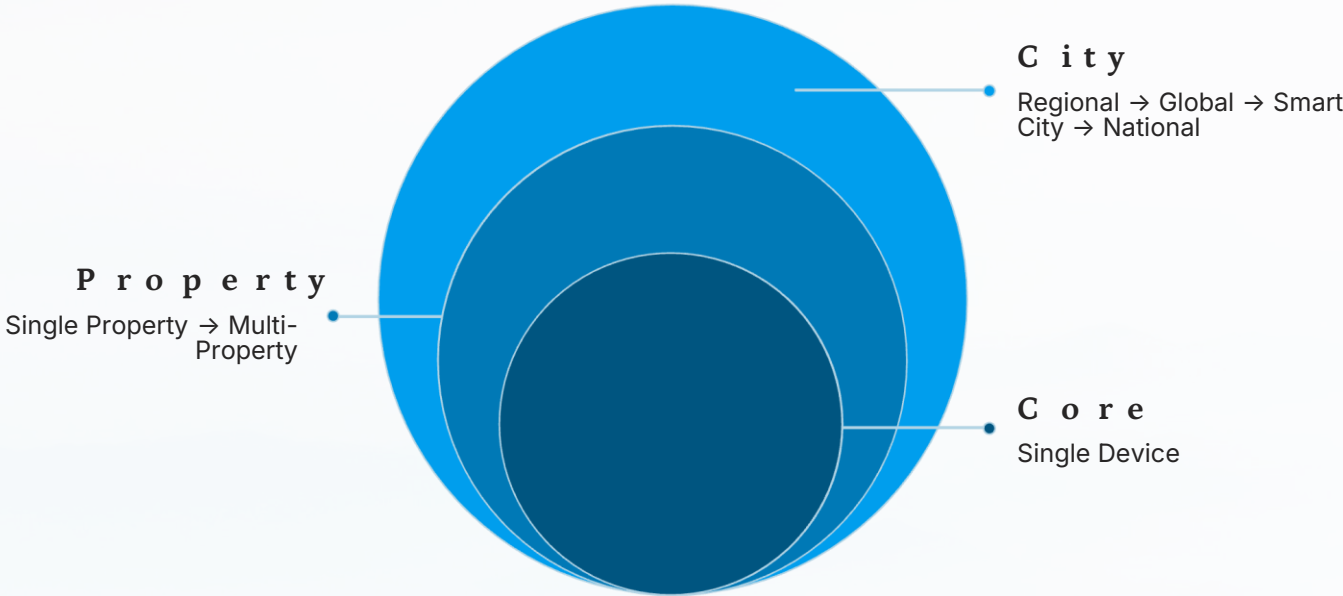
S e c u - F i r s t

- Security gates are **non-bypassable** at every stage
- No firmware ships without passing automated vulnerability scanning
- Container builds are signed and immutable
- OTA releases require cryptographic chain-of-trust validation

⚠ Zero exceptions — automated security enforcement is absolute, not advisory

Engineering Velocity Meets Enterprise Reliability

E n t e r p r i s e



M +

C o n n e c t e d

Millions of endpoints at full global scale

B +

E v e n t s

Billions of telemetry events processed daily

P B

D a t a

Petabyte-scale lakes — no redesign required to grow



Architecture is **horizontally scalable at every layer** — no redesign required to grow from single site to national infrastructure

Future



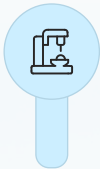
P h a s e C o n n e c t e d

Secure onboarding, real-time telemetry, remote management at scale



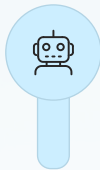
P h a s e D i g i t a l

Virtual asset models, simulation environments, predictive state management



P h a s e P r e d i c t i v e

ML-driven maintenance, energy optimization, and anomaly prediction



P h a s e A u t o n o m o u s

Self-healing systems, agentic operations, zero-touch management



P h a s e C o g n i t i v e

AI-orchestrated business outcomes across all platforms — CX, CK, CD, CE unified

Inspironics Architecture



Connectivity

WiFi 6E / Thread / BLE → Secure Gateways → MQTTS + TLS 1.3



Cloud

AWS / Azure / Google → Device Registry → Digital Twin → Time-Series Platform → AI/ML Engines → Agentic AI



Experience

Operations Portal → Mobile Platform → Business Outcomes across CX · CK · CD · CE

"From Factory Floor to Autonomous Enterprise"

